Gitesh Chawda

Machine Learning Engineer

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Summary

Software and Machine Learning Engineer with hands-on experience building and deploying production-grade backend and frontend systems across cloud environments. Experienced in designing and scaling LLM pipelines, agentic AI systems, and full-stack AI products with real-world deployment. Strong background in deep learning, computer vision, and transformer-based architectures with contributions across the Hugging Face and Keras ecosystems.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, SQL
ML and Deep Learning: PyTorch, TensorFlow, Keras 3, JAX, Hugging Face Transformers, scikit-learn, Transformers, Flash Attention, Diffusion Models, Computer Vision
LLMs, Inference, and Fine-tuning: vLLM, TensorRT, Triton, CUDA, XLA, bitsandbytes, Quantization, LoRA, PEFT, SFT, GRPO, KV Cache Optimization, Model Serving
Agentic AI Frameworks: LangChain, LangGraph, LlamaIndex, CrewAI, AutoGen, OpenAI Agents SDK, MCP, A2A
MLOps and Cloud: Docker, Linux, Git, GitHub Actions, CI/CD, MLOps, LLMOps, AWS, GCP
Web and Data: React.js, Flask, GraphQL, Node.js, NumPy, Pandas, OpenCV, MySQL, MongoDB, Oracle

Experience

KerasFormers, Author and Maintainer, Open Source (2024 - Present) Github

- Built a backend-agnostic, pip-installable Keras 3 library supporting 60+ architectures across vision, speech, LLM, and VLM tasks (SAM, DETR, DINOv3, Whisper, Llama, Qwen, DeepSeek, Mistral, Gemma, GLM), with single-codebase execution on TensorFlow, PyTorch, and JAX.
- Engineered Hugging Face-to-Keras checkpoint conversion with bit-close parity validation against reference implementations, and developed Keras 3 native weight-only quantization (INT8, INT4, FP8) using backend-agnostic operations, enabling up to 8x model size reduction for efficient inference deployment.

Open Source Contributor, Hugging Face Transformers, Keras, Keras.io, KerasCV (2022 - Present)

- Authored 42 merged PRs across Hugging Face Transformers, Keras, Keras.io, and KerasCV, improving model APIs, tokenizer design, type safety, and developer workflows.
- Added 10+ Keras 3 multi-backend APIs, layers, and utilities including PSNR, unravel_index, diagflat, RandomGrayscale, Equalization, and image augmentation workflows.
- Enhanced Hugging Face and Keras ecosystems through tokenizer modularity, type hints, ConvMixer models, YOLOv8 improvements, detection tests, segmentation augmentations, and official keras.io examples.

Fiserv, Software Developer, Chennai, India (July 2023 - August 2024)

- Developed and maintained a production web application with React.js, GraphQL, and AWS (AppSync, Lambda, API Gateway, Cognito, S3), supporting internal banking and payments workflows.
- Reduced unauthorized access attempts by 85% by strengthening authentication with AWS AppSync and Amazon Cognito, improving overall security posture.
- Secured fintech infrastructure by remediating 25+ critical vulnerabilities surfaced through internal audits and penetration testing, alongside the application security team.

Braynix AI, Deep Learning Intern, Bangalore, India (September 2021 - December 2021)

- Built deep learning pipelines for medical X-ray abnormality detection, improving validation accuracy by 20% over baseline CNN models across 5,000+ images.
- Reached 92% precision and 0.85+ IoU in medical image segmentation by implementing a U-Net with an EfficientNet-B4 encoder.
- Trained and benchmarked 3 object detection models (YOLOv5, SSD, RetinaNet) on 1,000+ labeled images at 0.78+ mAP.

Projects

txtcode: TypeScript npm package to orchestrate AI coding agents across 6 messaging platforms and 10 LLM backends with context-preserving switching. Github

kyolo: Keras 3 YOLO implementation with training, inference, pretrained weights, and multi-backend support. Github

Trinix: Triton-accelerated PyTorch optimization library with custom GPU kernels achieving up to 4.8x speedup. Github

OpenRag: Local-first RAG framework with switchable vector (Chroma) and knowledge-graph (Neo4j) retrieval, cross-encoder reranking, and local open-source LLMs (Ollama, Hugging Face) over a FastAPI and React stack. Github

Education

California State University, Fullerton, Master of Science in Computer Science, California, USA (August 2024 - May 2026)

Pimpri Chinchwad College of Engineering, Bachelor of Engineering in Computer Engineering, Pune, India (August 2019 - July 2023)

Publications

EffiConvRes: An Efficient Convolutional Neural Network with Residual Connections and Depthwise Convolutions, 2023 7th International Conference on Computing, Communication, Control and Automation (ICCUBEA).

Survey and Comparison of Various Pre-trained CNN Architectures and CNN-Transformer Combinations, Smart Grid and Electric Vehicle Conference.